

Business Requirements Document (BRD)

Project: Baseerah (بصيرة) – Early Warning System for Financial Health

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Document Status: Final Draft (MVP Expanded Scope)

1. Executive Summary

Baseerah (بصيرة) is an innovative, proactive AI-driven FinTech platform designed to revolutionize personal financial management and literacy in the Kingdom of Saudi Arabia. Unlike traditional reactive financial management applications that simply record past transactions after they occur, Baseerah acts as an **"Early Warning Financial Radar"**.

The platform aggregates advanced machine learning time-series forecasting, Generative AI (GenAI), and the Saudi Central Bank (SAMA) Open Banking Framework. It aims to predict liquidity gaps, cash flow deficits, and potential financial defaults 12 to 15 days before they happen. Furthermore, it introduces advanced **dynamic affordability simulation layers for users** and **predictive eligibility validation pipelines for banks**, transforming transactional data into predictive assets.

2. Strategic Objectives & Alignment with Saudi Vision 2030

- **Enhance National Saving Rates:** Directly supports the Saudi Vision 2030 strategic target of raising the average Saudi household savings rate from 6% to 10%.
- **Proactive Risk Management for Banks:** Empowers Alinma Bank to systematically reduce Non-Performing Loans (NPLs) and provisions by verifying affordability and capturing consumer credit strain before standard transactional defaults trigger.
- **Maximize Open Banking APIs Utilization:** Efficiently activates the newly deployed SAMA Open Banking infrastructure to serve as a smart, secure, and unified Multibank Aggregator.

- **Drive B2B2C Profitability (Win-Win Mechanism):** Displaces standard budgeting friction by turning personal financial insight into a high-converting cross-selling hub for the bank's own pre-approved micro-finance and micro-investment products.

3. MVP Scope (48-Hour Hackathon Deliverables)

3.1 In-Scope Deliverables:

- An intuitive and visually compelling front-end dashboard visualizing the daily **Financial Stress Score** via a responsive speedometer component.
- An asynchronous predictive backend model utilizing time-series analysis to calculate future cash flow trajectories and deficit timelines.
- A Generative AI (GenAI) conversational module supporting natural language processing for scenario simulations (e.g., automated lease or loan evaluations).
- **Interactive Loan Affordability Simulator:** Allows users to simulate custom loan amounts and specific bank interest rates to dynamically check their forward-looking repayment capability.
- **Bank Verifiable Credit Telemetry Reports:** Generation of predictive reports for the financial institution to confirm real-time transaction reliability and long-term loan capacity before/after product sign-offs.
- A simulated walkthrough of the **Smart Rescue Mode** demonstrating instantaneous micro-finance and micro-savings liquidation flows.

3.2 Out-of-Scope (Future Development Phases):

- Live, real-time API sync with the Saudi Credit Bureau (SIMAH).
- Direct cross-bank clearing and settlement engine architecture execution.

4. Functional Requirements

4.1 Financial Stress Score Dashboard

ID	Requirement Description	Priority
FR-01	The system shall ingest multibank transaction payloads via standardized APIs to calculate a daily Financial Stress Score (0-100) based on spending velocity, income consistency, and dynamic liability structures.	High
FR-02	The UI must express the calculated stress score dynamically through an absolute visual speedometer layout using distinct thresholds (Green = Optimal, Yellow = Warning, Red = Deficit Critical Risk).	High

4.2 GenAI Financial Scenario & Loan Affordability Simulation

ID	Requirement Description	Priority
FR-03	The platform shall enable users to input prospective, unstructured large purchasing actions in natural language or via invoice/quotation image uploads (e.g., "What happens if I sign a vehicle lease with a 2,100 SAR monthly payment?").	High
FR-04	The LLM engine must analyze historic seasonal variations (e.g., summer travel or tuition surges) and output a precise longitudinal prediction on how the action shifts liquidity profiles across 3, 6, and 12-month spans.	High
FR-05	[NEW] Loan Parameter Simulator: The system must allow users to define a prospective loan principal amount alongside the bank's specific interest rate (profit margin) to instantly verify dynamic installment load capabilities against their live income telemetry.	High

4.3 Smart Rescue Mode & Bank Cross-Selling Hub

ID	Requirement Description	Priority
FR-06	Upon predicting an institutional or consumer budget cash deficit, the application shall trigger a proactive push notification exactly 15 days before the expected payment failure to warn the user.	High
FR-07	The alert system must embed concrete, pre-approved action links allowing the user to bridge the shortfall: through an automated Micro-Finance (Murabaha) product or via safe asset liquidations from existing bank investment accounts.	High
FR-08	[NEW] B2B Credit Verification Pipeline: The system shall generate secure and automated data reports for the partner bank's underwriting teams, verifying whether a loan petitioner genuinely maintains the long-term cash flow stamina to sustain the debt without transitioning into non-performance status.	High

4.4 Gamified Micro-Saving Challenges

ID	Requirement Description	Priority
FR-09	The app shall deploy customized micro-saving objectives tailored to the individual's daily spending anomalies to systematically drive down their overall risk tiers.	Medium
FR-10	The system shall mock a rewards engine integrating directly with the hosting bank's existing loyalty ecosystem (e.g., Alinma Akthar Points) to reward compliant financial improvement.	Medium

5. Non-Functional Requirements

- **Security & Regulatory Compliance:** Synthetic data modeling and schema processing must strictly mimic SAMA Open Banking Framework tokenization constraints and NDMO local data rules. Bank reporting modules must ensure strict encryption layers.
- **Performance Standards:** The forecasting analytical framework must calculate data dumps and serve results under 2.5 seconds. Conversational AI responses must satisfy a Time-to-First-Token limit of less than 1.0 second.

- **Scalability Architecture:** Backend processes must reside on a stateless REST API layer built with Python FastAPI to cleanly support high concurrency requests during live jury demonstrations.

6. Core Technology Stack Selection

- **Frontend Framework:** Flutter (for native cross-platform deployment) or Next.js.
- **Backend Engine Architecture:** Python FastAPI for optimized execution and high-performance asynchronous execution.
- **Artificial Intelligence Ecosystem:** **Google Gemini API** for conversational contextual evaluation and logic translation, complemented by **Prophet / XGBoost** models for algorithmic time-series forecasting.
- **Sandbox Environment Data:** Bespoke mocked JSON datasets structurally matching active enterprise Open Banking payload telemetry formats.

Judging Advantage Strategy: Baseerah's distinct market superiority resides in its transition from consumer cost tracking to a highly monetization-centric risk reduction utility for Alinma Bank. By introducing automated financial reports for bank risk departments and an interactive custom loan simulator, it directly solves the bank's core challenge of underwriting accuracy—maximizing the score for the First Place prize.